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Bonus Project 4 - 4-Bit Binary Counter

The purpose of this project is to design and implement a 4-bit binary counter with speed setting.

Part 1: Parts

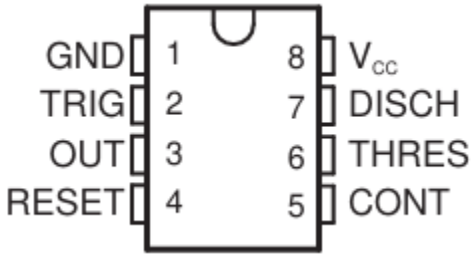
Parts used are listed with their product numbers:

Two JK Flip Flops [SN74HC112N] (or T type), I used 4 of them for visual purposes.

A potentiometer, preferably 100k [P0915N-FC15BR100K]. 555 type oscillator/timer [SA555P].

Part 2: Wiring / Diagram

Starting with the SA555P, its datasheet below. The wiring will be like this:



1: Ground
2: Connected to 6 (THRES)
3: Will go into the CLK of your first Flip Flop
4: 5V
5: 0.1uF capacitor to Ground (For filtering noise)
6: 10uF capacitor to Ground (May increase/decrease for adjusting blink speed)
7: 1K ohm to 5V (Base resistance Pre-Pot)
8: 5V

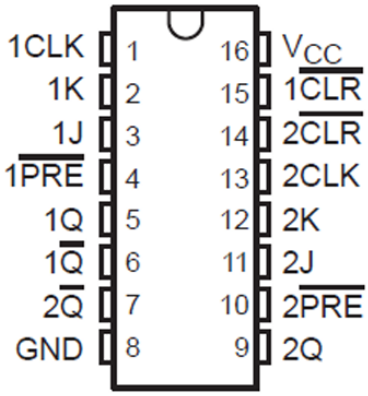
Then we have to connect our potentiometer into this IC unit. Potentiometer pins on your side:

Left: 5V
Middle: Pin 7 (DISCH) of SA555P
Right: Pin 6 (THRES) of SA555P

So some of the SA555P pins will have more than one connections to it:

Pin 3: 2 Connections: LED to Ground and CLK1 of your first flip flop
Pin 6: 3 Connections: Pin 2 SHORT, potentiometer right pin, 10uF capacitor to ground
Pin 7: 2 Connections: 1K Resistor to ground, Potentiometer middle pin

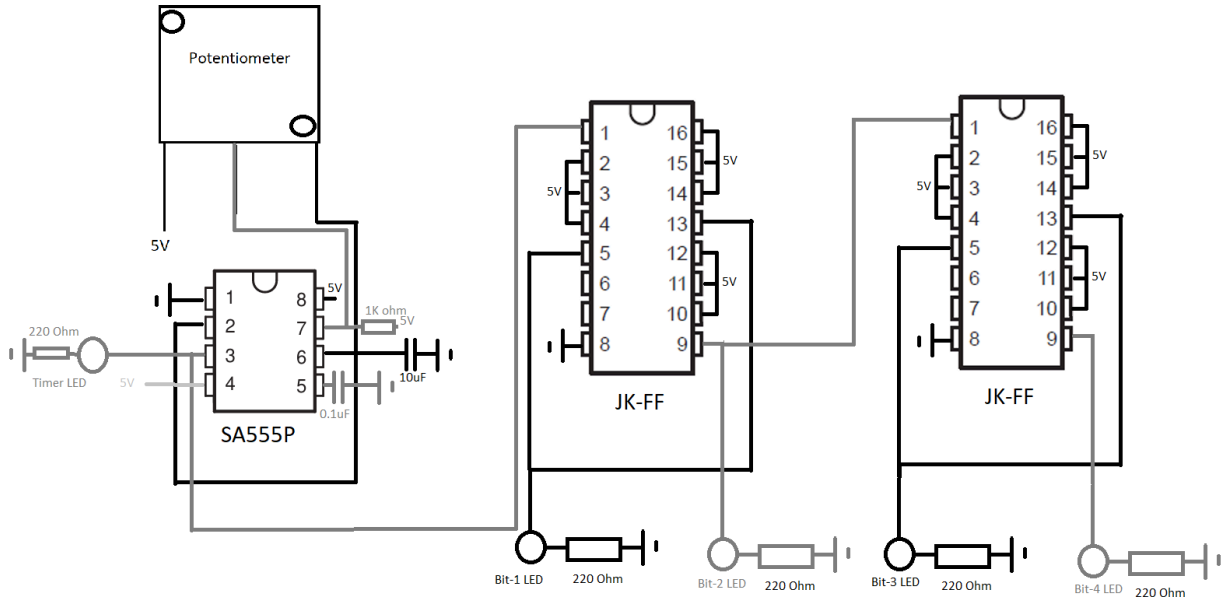
After the timer is set, we go on to wiring the flip-flops. JK Datasheet below.



Assuming both sides will be used:
1: SA555P Output (Pin 3) will go in here (If second chip, see [9])
2: 5V
3: 5V
4: 5V
5: Both an LED to the ground with 220 ohm. And into the CLK 2 on the other side.
6: Not needed
7: Not needed
8: Ground

9: Both an LED to the ground with 220 ohm. And into the CLK 1 of your next Chip
10: 5V
11: 5V
12: 5V
13: See [5]
14: 5V
15: 5V
16: 5V
You will repeat this one more time to get to 4 bit binary.

So the final diagram should look something like this:



Part 3: Working video example

https://tinyurl.com/52jjf5km

Conclusion

In conclusion, we were able to create a 4-bit binary counter using JK Flip-Flops with clock speed setting. The same can be done with a button instead to simulate manual CLK inputs, or extended upon with additional flip flops to go beyond 4 bits.